

Tuan Anh Nguyen (Ph.D.)

Director of Avionics

Assistant Professor

Dependable, Autonomous and Intelligent Systems (DAIS)



🏢 Director of Avionics, Viettel Aviation Technology Center (VATC), Viettel Group, Hanoi, Vietnam

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📖 Professional member of AIAA, IEEE, IEEE Computer Society, IEEE Robotics and Automation (IEEE RAS), IEEE Aerospace and Electronic Systems (IEEE AESS), and IEEE Reliability Societies

Brief summary paragraph

- "Director of Avionics in a Military Helicopter Project in Vietnam;
- Project Technical Leader and Lead Researcher of Artificial Intelligence Digital Twin Research for Smart Urban Air Mobility (UAM) (approx. KRW 7.7B over 9 years, 30 full-time M.Sc/Ph.D. researchers) funded by NRF in Korea.
- 5+ years teaching/supervision in US/Korea; 1+ year R&D on LLMs and LoRA; 5+ years in deep learning/generative AI (CNN, RNN/LSTM, GAN, Transformer, DRL: DQN/PPO/TRPO/A2C/A3C) and ML (surrogates, clustering) for digital twin dynamics/control, aerospace mission planning, and healthcare robotics. 12+ years AI cloud systems engineering focused on modeling, simulation, and dependability for aerospace/computing systems.
- Research focuses on Artificial intelligence, reinforcement learning and generative modeling for autonomous control, navigation, perception and optimization, AI-driven digital twins and dependable platform for advanced and urban air mobility and unmanned aerial systems; dependable cloud/edge/IoT platforms for autonomous and mission-critical systems; Quantitative dependability, performance and energy modeling of computing and network infrastructures, cyber-physical and robotic systems.
- Experienced in theory, simulation, and prototypes. Author of numerous top-tier publications. Contributions span AI, computing, and system engineering.

Experiences

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|-----------------------|---|
| Oct. 2025 – now | ◇ Director of Avionics , Viettel Aviation Technology Center, Viettel Group, Hanoi, Vietnam. |
| Jun. 2025 – Oct. 2025 | ◇ Principal Engineer , High-Tech Industry Division, Viettel Group; Hanoi, Vietnam. |
| Oct. 2024 – now | ◇ Lecturer , Software Engineering Department, Faculty of Information Technology, Ho Chi Minh City University of Industry and Trade (HUIT); Ho Chi Minh City, Vietnam. |
| Mar. 2021 – Jun. 2025 | ◇ Academic Research (Assistant) Professor , Konkuk University (KU)
(i) Department of Artificial Intelligence, Graduate School; (ii) Konkuk Aerospace Design-Airworthiness Institute (KADA); Seoul, South Korea. |
| Jun. 2020 – Feb. 2021 | ◇ Senior Research Associate , Konkuk University (KU), Konkuk Aerospace Design-Airworthiness Institute (KADA); Seoul, South Korea. |
| Mar. 2016 – Mar. 2020 | ◇ Assistant Professor , Konkuk University (KU), Office for Research of University-Industry Cooperation Foundation; Seoul, South Korea. |
| Aug. 2015 – Mar. 2016 | ◇ Postdoctoral Research Associate , Konkuk University (KU), Distributed Multimedia Systems Laboratory (DMS Lab.); Seoul, South Korea. |
| Mar. 2011 – Aug. 2015 | ◇ Graduate Researcher , Korea Aerospace University (KAU), Network Security and Embedded Systems Laboratory (NSLab.); Seoul, South Korea. |

Experiences (continued)

- Sep. 2009 – Feb. 2011 ♦ **Permanent Research Associate / Embedded Software Developer / Technical Team Leader**, FPT Technology Research Institute (FTRI), FPT Corporation, FSpace Laboratory; Hanoi, Vietnam.
- Aug. 2008 – Aug. 2009 ♦ **Embedded System Developer**, FPT Software Company, FPT Corporation, TRON Laboratory; Hanoi, Vietnam.
- Aug. 2008 – Aug. 2010 ♦ **Graduate Research Assistant**, Hanoi University of Science and Technology (HUST); Hanoi, Vietnam.

Education

- 2011 – 2015 ♦ **Ph.D., Computer Science and System Engineering**, Korea Aerospace University (KAU), Seoul, South Korea.
GPA: 4.0/4.0 (Full A); Advisors: Prof. Dr. Jong Sou Park (KAU), Prof. Dr. Dong Seong Kim (University of Queensland).
- 2008 – 2010 ♦ **M.Sc., Mechatronics**, Hanoi University of Science and Technology (HUST), Hanoi, Vietnam.
Thesis: Design and Analysis of F-1 pico-satellite System; Advisor: Prof. Dr. Dinh Van Phong.
- 2003 – 2008 ♦ **B.Eng, Mechatronics**, Hanoi University of Science and Technology (HUST), Hanoi, Vietnam.
Thesis: Simulation of Math-Mechanic Systems using Open-source Programming Library – Open Dynamics Engine (ODE); Advisor: Prof. Dr. Dinh Van Phong.

Projects

- Jun. 2020 – May 2029 ♦ **Artificial Intelligence Digital Twin Research for Smart Urban Air Mobility (UAM)**
Excellent Research Center Project funded by the National Research Foundation via the Ministry of Education Basic Science Program (approx. KRW 7.7B over 9 years; grant 2020R1A6A1A03046811).
Role: Project Technical Leader, Lead Researcher.
- Mar. 2022 – Feb. 2027 ♦ **Training Experts in Future Automobile R&D**
Professional Education Project funded by the Ministry of Trade, Industry and Energy (approx. KRW 7.7B over 9 years).
Role: Lead Researcher.
- Sep. 2021 – Aug. 2026 ♦ **Secure and Reliable Multi-Agent Public Safety Services for Disaster Tolerance in Urban Smart Cities (i-MeC2 Framework)**
Middle-Career Researcher Project supported by the National Research Foundation (Basic Science Program, Ministry of Education; approx. KRW 800M over 5 years; grant 2021R1A2C2094943).
Role: Lead Researcher.
- 2015 – 2020 ♦ **Distributed/Parallel Platform of Streaming Big Data Processing for Logistics and Transportation**
Information Technology Research Center; Institute for Information and Communications Technology Promotion; Ministry of Science, ICT and Future Planning, South Korea.
Role: Technical Staff.

Projects (continued)

- 2016 – 2019 ♦ **Developing High Performance Hybrid Cloud Computing Technologies for Knowledge Information Services and Human Resource Development**
Institute for Information and Communication Technology Promotion, South Korea.
Role: Technical Staff.
- 2017 – 2018 ♦ **International Cooperative Project Between Korea and Vietnam on Disaster-Tolerant Cloud Systems**
Institute for Information and Communication Technology Promotion, South Korea (approx. KRW 40M over 2 years), with Institute of Information Technology, VAST, Hanoi.
Role: Lead Researcher.
- 2014 – 2015 ♦ **Analysis Technique Study (ST-54) for Improving Integrated Survivability of Naval Ships**
Specialized Research Center for Survivability Technology supported by the Defense Acquisition Program Administration and the National Defense and Science Institute (contract UD120019OD), South Korea.
Role: Software Developer, Core Researcher.
- 2014 ♦ **Reliability Study of Advanced UAV**
National IT Industry Promotion Agency, South Korea.
Role: Researcher.
- 2011 – 2014 ♦ **UAV-Based Science and Technology Cooperation Between Korea and Peru for Resource Exploration and Environmental Monitoring**
National Research of Korea Project, South Korea.
Role: Core Software Developer, Technical Staff.
- 2013 ♦ **Balloon-Satellite Project for Training and Education**
Network Security and Embedded Systems Lab, Korea Aerospace University.
Role: Core Software/Hardware Systems Developer.
- ♦ **Construction of Historical Sites 3D Spatial Information Using Unmanned Airships**
Korea Cadastral Survey Corporation, South Korea.
Role: Researcher.
- 2012 ♦ **Unmanned Spacecraft Automatic Control System Development**
Korea Institute of Geoscience and Mineral Resources, South Korea.
Role: Researcher.
- 2009 – 2011 ♦ **F1 Pico-Satellite for Training and Education in Vietnam**
FPT Software, FPT Corporation, Vietnam.
Role: Core Software/Hardware Developer, Technical Subteam Leader.

Honors, Awards and Certificates

- Apr. 9, 2025 ♦ **Google UX Design**, Professional Certificate, Google (via Coursera). Completed 7 courses. Credential ID: ENMA187FFRQL.
Topics: UX design foundations; empathize, define, ideate; wireframes and low fidelity prototypes; UX research and early testing; high fidelity designs and prototyping in Figma; UI for websites; social good project and job preparation.
- ♦ **Google Advanced Data Analytics**, Professional Certificate, Google (via Coursera). Completed 7 courses. Credential ID: FDXM9D1LDJ1Q.
Topics: foundations of data science; Python; translating data into insights; statistics; regression; machine learning; capstone.

Honors, Awards and Certificates (continued)

- ◇ **Google Data Analytics**, Professional Certificate, Google (via Coursera). Completed 8 courses. Credential ID: EQFM48F55OA4.
Topics and tools: spreadsheets, SQL, Tableau, R; data preparation, cleaning, analysis; data visualization; capstone case study.
- ◇ **Google IT Support**, Professional Certificate, Google (via Coursera). Completed 5 courses. Credential ID: Q5CYJQFBEOE9.
Topics: troubleshooting and customer service; networking; operating systems; system administration and IT infrastructure services; IT security.
- ◇ **Google IT Automation with Python**, Professional Certificate, Google (via Coursera). Completed 6 courses. Credential ID: P8O2YQEH66PN.
Topics: Python; interacting with the operating system; Git and GitHub; troubleshooting and debugging; configuration management and cloud; automating real world tasks. Certificate statement includes passing graded assessments with score 80% or above.
- Apr. 7, 2025 ◇ **Google Project Management**, Professional Certificate, Google (via Coursera). Completed 6 courses. Credential ID: 3YRN2OFSORiL.
Topics: project initiation; planning; execution; agile project management; capstone applying project management in the real world.
- Apr. 9, 2025 ◇ **Google Business Intelligence**, Professional Certificate, Google (via Coursera). Completed 3 courses. Credential ID: oIRX1GUY15JK.
Topics: business intelligence foundations; data models and pipelines; dashboards and reports; building dashboards using design best practices for stakeholder needs.
- ◇ **Google Cybersecurity**, Professional Certificate, Google (via Coursera). Completed 8 courses. Credential ID: 4AJF8LNHKQOC.
Topics and tools: Python, Linux, SQL; SIEM tools; IDS; risk management; threats and vulnerabilities; detection and response; automation with Python; job preparation.
- ◇ **Google Digital Marketing and Ecommerce**, Professional Certificate, Google (via Coursera). Completed 7 courses. Credential ID: 8135OL52UXPR.
Topics: digital marketing and ecommerce foundations; customer engagement; social media leads; email marketing; marketing analytics and measurement; build and manage ecommerce stores; customer loyalty online.
- Mar. 2016 – Mar. 2020 ◇ **Professorship (KU Brain Pool Program for Talents)**, University-Industry Cooperation Foundation, Konkuk University (KU); Seoul, South Korea.
- Mar. 2016 – Mar. 2017 ◇ **Research Scientist Grant for Outstanding Young Researcher**, Office of Research, Konkuk University (KU); Seoul, South Korea.
- Mar. 2015 – Mar. 2016 ◇ **KU Postdoctoral Fellowship**, Konkuk University (KU); Seoul, South Korea.
- Dec. 2016 ◇ **First Author of Best Paper Award**, 11th KIPS International Conference on Ubiquitous Information Technologies and Applications (CUTE 2016); Bangkok, Thailand.
- 2011 – 2015 ◇ **Ph.D. Scholarship (USD 80,000)**, Korea Aerospace University (KAU); Seoul, South Korea.
- Feb. 14–21, 2014 ◇ **Representative Delegate of Republic of Korea (certificate)**, 4th ASEAN-Korea Frontier Forum (Asia Exchange Association; Ministry of Foreign Affairs; Association of Southeast Asian Nations); Seoul, South Korea.
- Jun. 5, 2014 ◇ **Certificate of Intellectual Properties Program**, Korea Invention Promotion Association (KIPA); Seoul, South Korea.

Honors, Awards and Certificates (continued)

- Jun. 23, 2013 ♦ **Certificate of Youth ICT Expert Training Program**, Korea Internet Security Agency (KISA) and Ministry of Science, ICT and Future Planning; Seoul, South Korea.
- 2008 ♦ **Certificate of Software Process for Developer (SPD) Program**, FPT Software, FPT Corporation; Hanoi, Vietnam.
- 2001 ♦ **First Prize in Physics Annual Excellent Student Contest**, Lien Ha High School; Hanoi, Vietnam.
- ♦ **First Prize in IT Annual Excellent Student Contest**, Dong Anh Gifted and Talented School; Hanoi, Vietnam.

Patents

- Jan. 2, 2024 ♦ **Apparatus and Method for Physics-Informed Generative AI with Transfer Learning Based Digital Twin Aerodynamics Development for eVTOL Aircraft**
Korea; application submitted.
- Jan. 3, 2024 ♦ **Apparatus and Method for Transition Flight Scheduling and Simulation of Tilt-Rotor eVTOL Aircraft Using Trim Analysis**
Korea; application submitted.
- Nov. 11, 2022 ♦ **Apparatus and Method for Performability Quantification of Switch-Over Moving Target Defense Mechanisms in Software Defined Network**
Korea Patent No. 10-2022-0150810.
- Dec. 22, 2021 ♦ **Apparatus and Method for Dependency and Security Quantification of IoMT System Based on Cloud-Fog-Edge Continuum Using Hierarchical Model**
Korea Patent No. 10-2021-0185171.

Software Registration

- Aug. 29, 2023 ♦ **Program for Predicting Aircraft Dynamics State Based on Deep Neural Network of Physical Information Using Flight Test Data**
Korea Software Registration No. C-2023-038352.
- ♦ **Program for Predicting Aircraft Dynamics State Based on Adversarial Generative Neural Network of Physical Information Using Flight Test Data**
Korea Software Registration No. C-2023-038351.
- ♦ **Program for Predicting Aircraft Dynamics State Based on Adversarial Generative Neural Network Using Flight Test Data**
Korea Software Registration No. C-2023-038350.
- Dec. 7, 2022 ♦ **Deep Learning-Based Program for Predicting Aircraft Dynamic State Using Flight Test Data**
Korea; registration filed.
- Nov. 30, 2022 ♦ **eVTOL Tiltrotor Aircraft Trim Analysis-Based Tilt Corridor Derivation Program**
Korea; registration filed.
- Nov. 25, 2022 ♦ **Robust Control Law es-DNLC Aircraft Attitude Control Program Based on PX4**
Korea; registration filed.
- Jun. 30, 2022 ♦ **Hierarchical Modeling and Prediction Program for Availability and Security Quantification of Industrial Internet of Things (IIoT)**
Korea Software Registration No. C-2022-026096.

Software Registration (continued)

- ◇ **Operational Cost and Availability Prediction Program for Virtualized Server Systems**
Korea Software Registration No. C-2022-026095.
- ◇ **Availability Prediction and Sensitivity Measurement Program for Smart Buildings Using Fog Computing**
Korea Software Registration No. C-2022-026094.
- Mar. 15, 2022 ◇ **Stochastic and Discrete-Event Simulation Program for Reliability, Availability, and Operational Cost Prediction of Software-Defined Networks**
Korea; registration filed.
- ◇ **Hierarchical Model-Based Program for Reliability and Availability Prediction of Cloud Data Center Networks**
Korea; registration filed.
- ◇ **Hierarchical Modeling and Prediction Program for Availability and Security Quantification of Internet of Industrial Things (IIoT)**
Korea; registration filed.
- ◇ **Availability Prediction and Sensitivity Measurement of Computing Infrastructures in Smart Buildings Aided by Fog Computing**
Korea; registration filed.
- Jan. 24, 2022 ◇ **Sensitivity Computation and Assessment Program for Business Continuity of Computing in Data Center Networks Relying on Server Virtualization**
Korea Software Registration No. C-2022-006597.
- ◇ **Verification Program for Operational Effectiveness of Disaster-Tolerant Technologies in Data Centers**
Korea Software Registration No. C-2022-006598.
- ◇ **Operational Availability and Cost Quantification Program for Virtualized Server Systems**
Korea Software Registration No. C-2022-006599.

Teaching

- Oct. 1, 2020 – Dec. 16, 2020 ◇ **Modern Artificial Intelligence Technologies**, Graduate School, Pacific States University; Los Angeles, CA, USA.
3424 Wilshire Blvd., 12th Floor.
Fall Quarter 2020 (11 weeks).
- Sep. 1, 2020 – Dec. 20, 2022 ◇ **AI-based Autonomous Systems**, Department of Artificial Intelligence, Graduate School, Konkuk University; Seoul, South Korea.
Fall Quarter 2022 (15 weeks).
- Mar. 1, 2024 – Jun. 20, 2024 ◇ **Vehicle Digital Twin and AI Technologies for Advanced Air Mobility**, Graduate School, Konkuk University; Seoul, South Korea.
Spring Quarter 2024 (14 weeks).

Supervision

- Mar. 2024 – May 2025 ◇ **Taemin Jeong**, M.Sc., DMS Lab., Konkuk University; Seoul, South Korea.
Enhanced Low-Rank Adapter Method for Accelerating Machine Learning with Large Language Models.

Supervision (continued)

- Mar. 2023 – May 2025 ◇ **Seungheop Nam**, M.Sc., DMS Lab., Konkuk University; Seoul, South Korea.
Efficient Boundary Selection-Based Robot Exploration: Deep Reinforcement Learning Combined with Heuristic Optimization.
- ◇ **Damsub Im**, Ph.D. Candidate, DMS Lab., Konkuk University; Seoul, South Korea.
Modeling and Analyzing Scalable Networked Data Storage Systems Using Random Reward Networks.
- Mar. 2021 – Dec. 2023 ◇ **Jeongseok Hyun**, M.Sc., KADA, Konkuk University; Seoul, South Korea.
eVTOL Tiltrotor Aircraft Transition Flight Scheduling and Simulation.
Currently, Research Engineer at Korea Aerospace Industries (KAI), South Korea.
- Mar. 2021 – Jun. 2023 ◇ **Min-Seok Jang**, M.Sc., KADA, Konkuk University; Seoul, South Korea.
Deep Learning based Flight Dynamics and Robust Control for High Precision eVTOL Vehicle Digital Twin.
Currently, Research Engineer at Korea Aerospace Industries (KAI), South Korea.
- ◇ **Tae-Ho Kwag**, M.Sc., KADA, Konkuk University; Seoul, South Korea.
Immersive Teleoperation using Physical-Digital Interaction of an eVTOL Digital Twin.
Currently, Research Engineer at Korea Aerospace Industries (KAI), South Korea.
- Aug. 2015 – Dec. 2017 ◇ **Luyao Zhou**, M.Sc., DMS Lab., Konkuk University; Seoul, South Korea.
Analysis and Simulation of a Small-Sized Cloud Computing System with Redundancy using Stochastic Reward Net and Cloudsim Framework.
Currently, Ph.D. candidate in Kyunghee University, South Korea.
- Aug. 2015 – Jun. 2016 ◇ **Ki-Hong Han**, M.Sc., DMS Lab., Konkuk University; Seoul, South Korea.
Stochastic Modeling and Discrete Event Simulation of Availability, Reliability and Security for Data Center Network.
Currently, first employment: back-end developer in Haezoom, Seoul, South Korea.
- Aug. 2016 – Aug. 2018 ◇ **Nang Lung Aung**, M.Sc., DMS Lab., Konkuk University; Seoul, South Korea.
Dependability Modeling System (DMS) Software Development for Computer Systems.
Currently, in Myanmar.

Education/Public Outreach Activities and Programs

- Sep. 2017 ◇ **Busan Global Smart Cities Challenger**, Young ICT Leader's Forum 2017, BEXCO, Busan, South Korea; hosted by Busan Metropolitan City (certificate).
- Aug. 9–12, 2017 ◇ Youth Week 2017 and International Youth 21 Conference 2017, hosted by UN-Habitat and YouthNow, Seoul, South Korea (certificate).
- Jul. 22–23, 2017 ◇ 3rd HackatSeoul — Smart Device for Seoul Bus, Seongsu Maker Space Center, Seoul, South Korea.
- Jul. 11–13, 2017 ◇ KF-EAI 5th Korea Friendship Program and KF Next Generation Conference 2017, hosted by Korea Foundation, Seoul, South Korea (certificate).
- Jan. 2015 ◇ SOKENDAI Asian Winter School 2015 — Eyes to Explore the Space Horizon, Sagami-hara Campus, JAXA, Kanagawa, Japan (visiting grant and paper presentation).
- May 2015 ◇ KF-ASEAN Korea Workshop, "Significance of Korea Unification for Peace and Prosperity of East Asia," Gangwon Province, South Korea (certificate of participation).

Education/Public Outreach Activities and Programs (continued)

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| Feb. 14–21, 2014 | ◇ 4th ASEAN-Korea Frontier Forum hosted by Asia Exchange Association and sponsored by Ministry of Foreign Affairs and Association of Southeast Asian Nations (certificate). |
| Mar. 31–Apr. 25, 2014 | ◇ 1st session of the Business Startup School offered by the Seoul Global Center (certificate). |
| Sep. 20, 2014 | ◇ 2014 Pan-Asian Student Council in Korea, organized by Korea Asian Friendship Foundation (certificate). |
| 2013 | ◇ Starting Trading Business in Korea Program, hosted by Korean International Trade Association (KITA) (certificate). |
| 2012 | ◇ Saemaul Undong Training Program for Foreign Students in Korea on social and infrastructure development in Saemaul Undong area and South Korea (certificate). |

Technical Skills

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| Programming Languages | ◇ C/C++/C#, VC++, VC#, Java, Python. |
| Mathematical Tools | ◇ Matlab, Maple, SHARPE, SPNP. |
| Coding Environment | ◇ PyTorch, TensorFlow, Linux-based, Windows-based, Embedded OS. |
| Hardware | ◇ Pixhawk (PX4), Arduino, mBed, Raspberry Pi. |

Ph.D. Dissertation Summary

Dissertation title: Availability Modeling and Analysis of Data Center Systems using Stochastic Models

Focus. My PhD research develops stochastic analytic models for availability prediction of computing systems including: (i) Virtualized Servers System (VSS), (ii) Virtualized Servers Network (VSN), (iii) Virtualized Data Center Network (vDCN), (iv) Software Defined Network (SDN), (v) Disaster Tolerant Data Center (DTDC), and (vi) Software Defined Data Center (SDDC).

Background. Virtualized computing systems in SDDCs and IaaS cloud environments are central to sustainable IT infrastructure, yet they must withstand security concerns, service interruptions, and reliability demands. Fault tolerance and disaster tolerance are essential to maintain high availability, price services accurately against Service Level Agreement (SLA) guarantees, and sustain continuity of enterprise information flows.

Key problems. Prior studies often simplify architectures to a single server, overlook complex dependencies among hardware and software components, and ignore the diversity of fault and disaster scenarios across multi-server virtualized systems. Such simplifications miss the realistic interactions and recovery behaviors needed for accurate assessment of practical systems.

Approach. I built comprehensive stochastic models with high-fidelity abstractions to capture multi-component failures, recoveries, and dependencies while keeping the models analytically tractable. Analyzing these models across availability metrics surfaces design insights and solution strategies for dependable virtualized systems.

Broad impact. The resulting models and availability assessments guide the design and analysis of contemporary systems, including IaaS cloud platforms, software defined data centers, and disaster tolerant data centers, enabling providers to plan redundancy, cost, and SLA commitments with quantitative confidence.

M.Sc. Thesis Summary

Thesis title: Design and Analysis of F-1 pico-Satellite System

Context. The cubesat concept has energized students and engineers worldwide by opening practical, small-scale access to space technology. Vietnam's FSpace Laboratory (FPT Software JSC. and FPT University) partnered with VAST to build the F-1 pico-satellite (10 cm × 10 cm × 10 cm, 1 kg) as a training platform and Earth observation mission, using Commercial Off-the-Shelf components to control cost and schedule.

System. F-1 carries two low-resolution cameras (0.3 MP), one high-resolution camera (1.0 MP), and sensors for temperature, magnetic field, and current to capture imagery and study the space environment. Critical subsystems – power supply unit (PSU), onboard computer (OBC), and communications (COM) – are double or triple redundant to ensure survivability.

Reliability focus. The thesis centers on designing and analyzing the satellite for high reliability against launch vibrations and harsh on-orbit thermal and radiation conditions. Results show the design is robust to key risks, enabling continuous long-term operation and forming the basis for subsequent manufacturing and test phases.

Research Publications

Last Update: Feb. 4, 2026. For more, please visit <https://anhnt2407.github.io/publications/>.

A. Manuscripts under review / revision

- Nam, S., Nguyen, T. A., Choi, E., & Min, D. **FH-DRL: Exponential-Hyperbolic Frontier Heuristics with DRL for Accelerated Exploration in Unknown Environments.** *IEEE Transactions on Intelligent Vehicles* (Manuscript ID: T-IV-25-02-0130; submitted 14-Feb-2025).
- Lee, U., Hwang, C., Park, Y., Jin, H., Nguyen, T. A., Choi, E., & Min, D. **VAE+DDPG: An Attention-Enhanced Variational Autoencoder for Deep Reinforcement Learning-Based Autonomous Navigation in Low-Light Environments.** *Advanced Intelligent Systems* (Manuscript No.: aisy.202500636R2; R2).
- Nguyen, T. A., Lim, D., Kyung, M., & Min, D. **Distributed Edge Storage Systems: Proactive High-Availability Microservices with Live Migration and Rejuvenation Strategies.** Submitted to *Mathematics* (MDPI).
- Nam, S., Nguyen, T. A., Choi, E., & Min, D. **Multi-head Fusion-based Actor-Critic Deep Reinforcement Learning with Memory Contextualisation for End-to-End Autonomous Navigation.** Manuscript under review.
- Folåsen, H., Nguyen, T. A., & Min, D. **FreqStack: Adaptive Spectral Mixture-of-Experts for Robust Zero-shot Deepfake Detection.** Manuscript under review.
- Folåsen, H., Min, D., & Nguyen, T. A. **From Global FFT to Local DWT: A Soft MoE Detector for Robust Deepfake Recognition.** Manuscript under review.

B. Domain 1. Artificial intelligence and reinforcement learning for autonomous control, navigation and perception

Deep and multi-agent reinforcement learning, neural control, physics-informed and generative models, and work with vision language and large language models for robustness, security, and control.

Journal and book publications

1. [P04] Park, J., Nguyen, T. A., & Min, D. (2025). **RT-VLM: Re-Thinking Vision Language Model with 4-Clues for Real-World Object Recognition Robustness**. *arXiv preprint*, arXiv:2509.05333. doi:10.48550/arXiv.2509.05333. [2025-09-01]
2. [P03] Nam, S., Nguyen, T. A., Choi, E., & Min, D. (2025). **Multi-head Fusion-based Actor-Critic Deep Reinforcement Learning with Memory Contextualisation for End-to-End Autonomous Navigation**. doi:10.36227/techrxiv.173747402.29145488/v1. [2025-01-21]
3. [P02] Nam, S., Nguyen, T. A., Choi, E., & Min, D. (2025). **FH-DRL: Exponential-Hyperbolic Frontier Heuristics with DRL for accelerated Exploration in Unknown Environments**. *arXiv preprint*, arXiv:2407.18892. doi:10.48550/arXiv.2407.18892. [2024-07-27]
4. [J25] Jeon, S., Lee, H., Kaliappan, V. K., Nguyen, T. A., Jo, H., Cho, H., & Min, D. (2022). **Multiagent Reinforcement Learning Based on Fusion-Multiactor-Attention-Critic for Multiple-Unmanned-Aerial-Vehicle Navigation Control**. *Energies*, 15(19), 7426. doi:10.3390/en15197426. (SCIE, JCR.Q3 Energy/Fuels., Ranking 80/119, IF2021=3.252) [2022-10-10]
5. [J22] Battisti, F., Silva, A., Pereira, L., Carvalho, T., Araujo, J., Choi, E., Nguyen, T. A., & Min, D. (2022). **hLSTM-Aging: A Hybrid LSTM Model for Software Aging Forecast**. *Applied Sciences*, 12(13). doi:10.3390/app12136412. (SCIE, JCR.Q2 Engineering, 32/91, IF2020=2.679) [2022-06-24]
6. [J20] Kaliappan, V. K., Nguyen, T. A., Min, D., Lee, J.-W., & Sakthi, U. (2022). **Steering Behavior-based Multiple RUAV Obstacle Avoidance Control**. *Intelligent Automation & Soft Computing*, 34(1), 575–591. doi:10.32604/iasc.2022.024577. (SCIE, JCR.Q3 Automation & Control., Ranking 44/64, IF2020=1.647) [2022-01-01]
7. [J09] Maw, A. A., Tyan, M., Nguyen, T. A., & Lee, J.-W. (2021). **iADA*-RL: Anytime Graph-Based Path Planning with Deep Reinforcement Learning for an Autonomous UAV**. *Applied Sciences*, 11(9). doi:10.3390/app11093948. (SCIE, JCR.Q2 Engineering, 32/91, IF2019=2.474) [2021-04-27]

Conference and preprint publications

8. [C44] Nam, S., Nguyen, T. A., Kang, S., & Min, D. (2025). **PGELU: Metaverse-Driven Deep Learning Boosted by a Parametric GELU for Stable and Scalable Recognition of Emotions and 3D Objects**. In *Proceedings of the 2025 International Conference on Metaverse Computing, Networking and Applications (MetaCom)*. doi:10.1109/metacom65502.2025.00044. [2025-08-27]
9. [C43] Park, H., Min, D., Nguyen, T. A., & Choi, E. (2025). **Adaptive Prompting in the Metaverse: An Iterative Prompt Optimization Framework for Enhancing LLM Performance Across Diverse Tasks**. In *Proceedings of the 2025 International Conference on Metaverse Computing, Networking and Applications (MetaCom)*. doi:10.1109/metacom65502.2025.00009. [2025-08-27]
10. [C45] Park, S., Min, D., Nguyen, T. A., & Choi, E. (2025). **Metaverse Security: LLM-Based Malicious Code Detection Through Token Optimization**. In *Proceedings of the 2025 International Conference on Metaverse Computing, Networking and Applications (MetaCom)*. doi:10.1109/metacom65502.2025.00057. [2025-08-27]
11. [C42] Kang, H., Lee, U., Park, Y., Min, D., Nguyen, T. A., & Choi, E. (2025). **Metaverse-Ready Simulation: Advancing Sim-to-Real by Reinforcement Learning for TurtleBot in ROS2 and Unreal Engine**. In *Proceedings of the 2025 International Conference on Metaverse Computing, Networking and Applications (MetaCom)*. doi:10.1109/metacom65502.2025.00008. [2025-08-27]
12. [P05] Jung, T., Seo, W., Park, H., Park, S., Nguyen, T. A., & Min, D. (2024). **ULoRA: Universal Low-Rank Adaptation of Diverse Deep Learning Architectures**. *ICLR 2025 Conference, withdrawn submission*. openreview.net/forum?id=bYsieh8LE2. [2024-10-04]
13. [C48] Nguyen, T. A., Jang, M., Hyun, J., Kwak, T., Toe, M., & Lee, J.-W. (2023). **Can Large Language Models Stabilize the Attitude of a UAM Vehicle? - A Preliminary Research**. In *Proceedings of the Korean Society for Aeronautical & Space Sciences Fall Conference 2023*, 1411–1412. [2023-11-15]
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D. Domain 3. IoT, smart environments and cyber physical systems

Internet of Things, smart cities and buildings, healthcare and industrial IoT, IoRT, and IoT-based disaster detection.

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E. Domain 4. Digital twin and dependable platforms for advanced and urban air mobility and unmanned aerial systems

Digital twins for UAM and AAM vehicles, cloud-in-the-loop platforms, software aging in digital twin backends, and mission dependability of UAS.

Journal and book publications

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F. Domain 5. Aerospace and space systems Aerodynamic modeling and data fusion, airfoil design with learning models, eVTOL flight mechanics and control, and early work on unmanned airships and nano satellites.

Journal and book publications

100. [J33] Pham, V., Tyan, M., Nguyen, T. A., & Lee, J.-W. (2023). **Extended Hierarchical Kriging Method for Aerodynamic Model Generation Incorporating Multiple Low-Fidelity Datasets**. *Aerospace*, 11(1), 6. doi:10.3390/aerospace11010006. (SCIE, JCR.Q1 Aerospace, Ranking 8/34, IF2022=2.6) [2023-12-20]
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103. [C50] Pham, V., Nguyen, N., Nguyen, T. A., & Lee, J.-W. (2024). **Multi-fidelity Data Fusion Method for Real-time Aerodynamic Modeling from Flight Data**. In *Proceedings of the Korean Society for Aeronautical & Space Sciences Fall Conference 2024*, 89–90. [2024-11-13]
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105. [C32] Hyun, J., Jang, M., Nguyen, T. A., & Lee, J.-W. (2023). **Transition Control Planning and Optimization for a Boxed-wing eVTOL Tiltrotor Vehicle using Trim Analysis**. In: *2023 IEEE International Conference on Unmanned Aircraft Systems (ICUAS)*, 1128–1135. doi:10.1109/ICUAS57906.2023.10156619. [2023-06-06]
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Academic Services

Invited Speaker/Presentation

- | | |
|----------------------|---|
| Oct. 17–20, 2023 | ◇ IEEE ICCAS 2023 , Yeosu, Korea.
23rd IEEE International Conference on Control, Automation and Systems;
2023.iccas.org.
Oral presentation: <i>PIGD-TL: Physics-Informed Generative Dynamics with Transfer Learning</i> . |
| Oct. 11–13, 2023 | ◇ Elsevier AFCS 2023 , Ho Chi Minh City, Vietnam.
Aviation Future: Challenges and Solutions; afcs.vn.
Oral presentation: <i>PIND-UAM: Physics-Informed Neural Dynamics of Boxed-Wing eVTOL Aircraft for UAM Vehicle Digital Twin</i> . |
| Jul. 16, 2023 | ◇ ACVYS 2023 , Sejong University, Seoul, South Korea.
The 9th Annual Conference of Vietnamese Young Scientists; acvys.org.
Oral presentation: <i>Digital Twin Technology for Next Generation Urban Air Mobility (UAM-ODT)</i> . |
| May 6–9, 2023 | ◇ IEEE ICACI 2023 , Seoul, Korea.
The 15th International Conference on Advanced Computational Intelligence; icaci2023.com.
Oral presentation: <i>QR-GAN: Generative Adversarial Networks meet Quantile Regression</i> . |
| Oct. 20–21, 2022 | ◇ IEEE Cloud Summit 2022 , George Mason University, Fairfax, Virginia, USA.
ieeeccloudsummit.org.
Oral presentation: <i>Survivability and Resiliency Modeling and Analysis of an Internet of Industrial Things using Hierarchical Models</i> . |
| Jun. 27–Jul. 1, 2022 | ◇ AIAA Aviation Forum , Chicago, IL, USA.
aiaa.org/aviation.
Oral presentation: <i>Model-driven Mission Dependability Design of Unmanned Aerial Systems</i> . |
| Sep. 4, 2017 | ◇ Springer INISCOM 2017 , Ho Chi Minh City, Vietnam.
3rd EAI International Conference on Industrial Networks and Intelligent Systems; archive.iniscom.org/2017.
Oral presentation: <i>Model-based Sensitivity of a Disaster Tolerant Active-Active GENESIS Cloud System</i> . |
| Jun. 1, 2017 | ◇ ACVYS 2017 , Seoul, South Korea.
Annual Conference of Vietnamese Young Scientists in Korea; acvys.org.
Invited speaker and ICT Session Chair: <i>Smart City in the 4th Industry Revolution (4IR)</i> . |
| Dec. 19–21, 2016 | ◇ Springer CUTE 2016 , Bangkok, Thailand.
(Springer) 11th KIPS International Conference on Ubiquitous Information Technologies and Applications; cute-conference.org/2016.
Oral presentation: <i>An Evaluation of Availability, Reliability and Power Consumption for a SDN Infrastructure using Stochastic Reward Net</i> . |

Academic Services (continued)

- May 1, 2016 ◇ **ACVYS 2016**, Seoul, South Korea.
Annual Conference of Vietnamese Young Scientists in Korea; acvys.org.
Invited speaker and ICT Session Chair.
- Feb. 26–27, 2014 ◇ **ICCA 2014**, Yangon, Myanmar.
12th International Conference on Computer Applications;
ucsy.edu.mm/onlineresource/handle/123456789/107.
Oral presentation: *Availability Modeling and Analysis in a Virtualized Servers Network*.
- Jun. 2012 ◇ **IEEE ICES 2012**, Liverpool, UK.
9th IEEE International Conference on Embedded Software and Systems; ieeeexplore.ieee.org/document/6332394.
Oral presentation: *Design and Implementation of Embedded Hardware and Software Architecture in an Unmanned Airship*.

Journal Review

- Adv. Eng. Inform. ◇ **Advanced Engineering Informatics** (Elsevier; SCIE/Scopus); journals.elsevier.com/advanced-engineering-informatics. Data- and AI-driven methods for engineering design and construction.
- Adv. Space Res. ◇ **Advances in Space Research** (Elsevier; SCIE/Scopus); journals.elsevier.com/advances-in-space-research. Space science, earth observation, planetary and atmospheric research.
- AST ◇ **Aerospace Science and Technology** (Elsevier; SCIE/Scopus); sciencedirect.com/journal/aerospace-science-and-technology. Aeronautics and astronautics structures, control, and propulsion.
- Applied Energy ◇ **Applied Energy** (Elsevier; SCIE/Scopus); sciencedirect.com/journal/applied-energy. Energy systems, renewables, decarbonization, and sustainability.
- CAAI Trans. IT ◇ **CAAI Transactions on Intelligence Technology** (CAAI/Elsevier; SCIE/Scopus); sciencedirect.com/journal/caai-transactions-on-intelligence-technology. Artificial intelligence, robotics, vision, and language.
- Computing ◇ **Computing** (Springer; SCIE/Scopus); springer.com/journal/607. Computer science theory and systems, algorithms, and applications.
- Discover Computing ◇ **Discover Computing** (Springer Nature; open access); link.springer.com/journal/44217. Computing foundations, AI, data, and applications.
- EAAI ◇ **Engineering Applications of Artificial Intelligence** (Elsevier; SCIE/Scopus); sciencedirect.com/journal/engineering-applications-of-artificial-intelligence. Applied AI for engineering, manufacturing, and cyber-physical systems.
- Frontiers Earth Sci. ◇ **Frontiers in Earth Science** (Frontiers; SCIE/Scopus); frontiersin.org/journals/earth-science. Geoscience, geophysics, climate, and planetary science.
- FGCS ◇ **Future Generation Computer Systems** (Elsevier; SCIE/Scopus); sciencedirect.com/journal/future-generation-computer-systems. Distributed, cloud/edge, HPC, and dependable computing.
- IEEE Access ◇ **IEEE Access** (IEEE; SCIE/Scopus); ieeeaccess.ieee.org. Multidisciplinary engineering and computing; regular reviewer.
- IEEE IoT-J ◇ **IEEE Internet of Things Journal** (IEEE; SCIE/Scopus); comsoc.org/publications/journals/ieee-iot-journal. IoT, CPS, sensing, and networked systems.

Academic Services (continued)

IEEE Systems J.	◇ IEEE Systems Journal (IEEE Systems Council; SCIE/Scopus); ieeesystem-sjournal.org . Complex systems engineering, integration, and dependability.
IEEE TII	◇ IEEE Transactions on Industrial Informatics (IEEE IES; SCIE/Scopus); ieee-ies.org/pubs/transactions-on-industrial-informatics . Industrial IoT, CPS, automation, and analytics.
IEEE TCAS-I	◇ IEEE Transactions on Circuits and Systems I (IEEE CAS; SCIE/Scopus); ieee-cas.org/publication/tcas1 . Analog/digital circuits, signal processing, and systems.
IEEE TCC	◇ IEEE Transactions on Cloud Computing (IEEE CS; SCIE/Scopus); computer.org/csdl/journal/cc . Cloud architecture, security, virtualization, and services.
IEEE T-Cyb	◇ IEEE Transactions on Cybernetics (IEEE SMC; SCIE/Scopus); ieeesmc.org/publications/transactions-on-cybernetics . Control, learning, and intelligent systems.
IEEE TETC	◇ IEEE Transactions on Emerging Topics in Computing (IEEE CS; SCIE/Scopus); computer.org/csdl/journal/ec . Novel computing paradigms, architectures, and systems.
IEEE TNSM	◇ IEEE Transactions on Network and Service Management (IEEE ComSoc; SCIE/Scopus); comsoc.org/publications/journals/ieee-tnsm . Network/service management, SDN/NFV, orchestration.
IEEE T-SusC	◇ IEEE Transactions on Sustainable Computing (IEEE CS; SCIE/Scopus); computer.org/csdl/journal/st . Green/energy-efficient computing and systems.
IJASS	◇ International Journal of Aeronautical and Space Sciences (Springer; SCIE/Scopus); springer.com/journal/42405 . Aeronautics, astronautics, and flight systems.
Int. J. Aerospace Eng.	◇ International Journal of Aerospace Engineering (Hindawi; SCIE/Scopus); hindawi.com/journals/ijae . Design, control, propulsion, and materials for aerospace.
Wiley IJCS	◇ International Journal of Communication Systems (Wiley; SCIE/Scopus); onlinelibrary.wiley.com/journal/10991131 . Network protocols, wireless, and systems.
IoT & CPS	◇ Internet of Things and Cyber-Physical Systems (KeAi/Elsevier; Scopus); sciencedirect.com/journal/internet-of-things-and-cyber-physical-systems . IoT, CPS, industrial analytics.
JCOMP	◇ Journal of Computational Physics (Elsevier; SCIE/Scopus); sciencedirect.com/journal/journal-of-computational-physics . Numerical methods, PDEs, and scientific computing.
JIII	◇ Journal of Industrial Information Integration (Elsevier; SCIE/Scopus); sciencedirect.com/journal/journal-of-industrial-information-integration . Industrial data integration, IoT/edge/cloud, and analytics.
IntelCom	◇ Journal of Information and Intelligence (KeAi/Elsevier; Scopus); sciencedirect.com/journal/journal-of-information-and-intelligence . Information science, AI, and data-centric intelligence.
JKSUCIS	◇ Journal of King Saud University – Computer and Information Sciences (Elsevier; SCIE/Scopus); sciencedirect.com/journal/journal-of-king-saud-university-computer-and-information-sciences . Computer science, AI, and data engineering.

Academic Services (continued)

JOI-TMC	◇ Journal of Open Innovation: Technology, Market, and Complexity (MDPI; SCIE/Scopus); mdpi.com/journal/joitmc . Open innovation, technology management, and entrepreneurship.
RESS	◇ Reliability Engineering and System Safety (Elsevier; SCIE/Scopus); sciencedirect.com/journal/reliability-engineering-and-system-safety . System safety, risk, dependability, and resilience.
SPE	◇ Software: Practice and Experience (Wiley; SCIE/Scopus); onlinelibrary.wiley.com/journal/1097024x . Empirical software engineering, systems, and tools.
SUPE	◇ The Journal of Supercomputing (Springer; SCIE/Scopus); link.springer.com/journal/11227 . High-performance computing, parallel algorithms, and applications.
JSA	◇ Journal of Systems Architecture (Elsevier; SCIE/Scopus); sciencedirect.com/journal/journal-of-systems-architecture . Computer architecture, embedded and real-time systems.
Measurement	◇ Measurement (Elsevier; SCIE/Scopus); sciencedirect.com/journal/measurement . Sensors, instrumentation, and metrology.
Ocean Eng.	◇ Ocean Engineering (Elsevier; SCIE/Scopus); sciencedirect.com/journal/ocean-engineering . Marine/offshore engineering, hydrodynamics, and structures.
PLOS ONE	◇ PLOS ONE (PLOS; SCIE/Scopus); journals.plos.org/plosone . Multidisciplinary open-access science and engineering.
Results Control Optim.	◇ Results in Control and Optimization (Elsevier; Scopus); sciencedirect.com/journal/results-in-control-and-optimization . Control theory, optimization, reinforcement learning.
Scientific Reports	◇ Scientific Reports (Nature Portfolio; SCIE/Scopus); nature.com/srep . Multidisciplinary open-access science and engineering.
SUSCOM	◇ Sustainable Computing: Informatics and Systems (Elsevier; SCIE/Scopus); sciencedirect.com/journal/sustainable-computing-informatics-and-systems . Green IT, energy-aware systems, and sustainability analytics.
TSRI	◇ Sustainable and Resilient Infrastructure (Taylor & Francis; Scopus); tandfonline.com/journals/tsri20 . Resilient infrastructure, risk, and adaptive design.

Reviewer/TPC of Conferences

AAAI 2026	◇ AAAI Conference on Artificial Intelligence 2026 ; top-tier AI research venue (AAAI/AAAI Press); aaai.org/conference/aaai/ . Reviewer.
AISTATS 2026	◇ International Conference on Artificial Intelligence and Statistics 2026 ; leading ML/statistics conference (PMLR proceedings); aistats.org . Reviewer.
ICLR 2025	◇ International Conference on Learning Representations 2025 ; premier deep learning venue with open review; iclr.cc . Reviewer.
ICAIT 2024	◇ International Conference on Artificial Intelligence Technology 2024 ; AI/ML systems and applications; icait.org . Reviewer/TPC.
AIAA Aviation 2021	◇ AIAA Aviation Forum 2021 ; American Institute of Aeronautics and Astronautics flagship forum for aviation and aerospace applications; aiaa.org/aviation . Reviewer/TPC.
CECNet 2017	◇ The 7th International Conference on Electronics, Communications and Networks ; cecnetconf.org . Reviewer.

Academic Services (continued)

ICACCI 2017	◇ IEEE International Conference on Advances in Computing, Communications and Informatics 2017 ; icacci-conference.org/2017 . Reviewer.
ICACCI 2016	◇ IEEE International Conference on Advances in Computing, Communications and Informatics 2016 (Poster Track); icacci-conference.org/2016 . Reviewer.
ISSPIT 2015	◇ IEEE International Symposium on Signal Processing and Information Technology 2015 ; ece.adu.ac.ae/ISSPIT2015 . Reviewer.
CoCoNet 2015	◇ IEEE International Conference on Computing and Network Communications 2015 (Poster Track); iiitmk.ac.in/coconet2015/poster.html . Reviewer.
RASAS 2014	◇ International Workshop on Reconfigurable Architectures and Self-adaptive Autonomic Systems 2014 ; icacci-conference.org/website/rasas2014 . Reviewer.
IBMSGs 2015	◇ International Biometrics & Smart Government Summit 2015 ; ibmsgs.org . Reviewer.
TrustCom 2015	◇ IEEE International Conference on Trust, Security and Privacy in Computing and Communications 2015 ; research.comnet.aalto.fi/Trustcom2015 . Reviewer.
ICCAAM 2014	◇ International Conference on Control, Automation and Applied Mechanics 2014 ; siggraph.org/event/iccaam-2014 . Reviewer.

References and Collaborators

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